

VASAVI COLLEGE OF ENGINEERING (AUTONOMOUS)

IBRAHIMBAGH, HYDERABAD-31

Department of Computer Science and Engineering

BE-III SEMESTER I-Internal Examination

Name of the Course: **Data Structures**

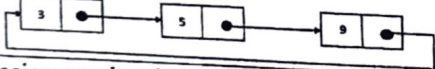
Course Code: UI21PC320CS and UII21PC320CS

Max Marks: 30

Date: 13.11.2023

Time: 2.30 to 04.00 PM

Note: Answer All the questions

Q.No.	Description of the question	Mar ks	BTL (1/2/3/ 4/5/6)	Mapped																	
				CO	PO																
Part-A (6x 1=6 Marks)																					
1. ✓	<table border="1"><tr><td>4</td><td>0</td><td>0</td><td>0</td></tr><tr><td>4</td><td>4</td><td>0</td><td>0</td></tr><tr><td>4</td><td>4</td><td>7</td><td>0</td></tr><tr><td>7</td><td>9</td><td>6</td><td>8</td></tr></table> What kind of matrix is this. How much memory is required to store it efficiently.	4	0	0	0	4	4	0	0	4	4	7	0	7	9	6	8	1	1	1	1,2
4	0	0	0																		
4	4	0	0																		
4	4	7	0																		
7	9	6	8																		
2. ✓	Represent the Given Polynomial expression efficiently using arrays and using Linked List $2x^8+4x^6+3x^5+4x^3+8$	1	1	1	1,2																
3. ✓	Find the time complexity of the following code segment. <pre>for (c = 0; c < m; c++) { for (d = 0; d < q; d++) { sum = sum + first[c][d] } }</pre>	1	3	1	1,2																
4. ✓	Find the value of the following postfix expression $7\ 40\ 5\ /\ 4\ *\ 2\ /\ +$	1	3	2	1,2																
5. ✓	What kind of list is this. Design a function to insert the node at the end of this type of list 	1	3	2	1,2																
6. ✓	Design an algorithm to count the number of elements in the circular queue.	1	1	2	1,2																
Part-B (4x 3=12 Marks)																					
7. ✓	Given an array of random numbers, Push all the zero's of a given array to the end of the array. For example, if the given arrays is {1, 9, 8, 4, 0, 0, 2, 7, 0, 6, 0}, it should be changed to {1, 9, 8, 4, 2, 7, 6, 0, 0, 0, 0}. The order of all other elements should be same. Expected time complexity is O(n) and extra space is O(1). Input : arr[] = {1, 2, 0, 4, 3, 0, 5, 0}; Output : arr[] = {1, 2, 4, 3, 5, 0, 0, 0};	3	1	1	1,2																
8. ✓	Given a lists of students data of 4/4/ CSE and 4/4ECE department separately in descending order of marks for the placement . Write a function to create the single list which includes data from both departments in the descending order of marks.	3	3	1	1,2																

9.	Given an expression string exp , write a program to examine whether the pairs and the orders of "{", "}", "(", ")", "[", "]" are correct in the given expression. Input: exp = "[(){}{}()()]" Output: Balanced Input: exp = "[()]" Output: Not Balanced	3	2	2	1,2
10.	A project expo contest is being organized in a college as part of Engineers Day Celebration. The coordinators are supposed to keep track of student details such as roll number, name, class and branch who are shortlisted in the first round. Design a C program to store the student details in a linked list and count the number of students participated from CSE and IT department.	3	3	2	1,2
Part-C (2x 6=12 Marks)					
11a)	Given a 2D array, print it in spiral form. Input: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 Output: 1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10	3	2	1	1,2
b)	Design a function to add two polynomials when polynomials are stored using arrays SUM OF $10X^5+20X^4+5X$ $20X^8+10X^6+4X$ IS $20X^8+10X^6+10X^5+20X^4+9X$	3	3	1	1,2
12a)	Design an algorithm for infix to postfix expression conversion, showing the stack content and output expression after each operator read from the given input expression. $(1-2/3)*(4-5)+6/7*3$	3	2	2	1,2
b)	Write a function to reverse a linked list Input: Head of following linked list 1->2->3->4->5->NULL Output: Linked list should be changed to, 5->4->3->2->1->NULL	3	3	2	1,2

Summary of the percentage for each of the criteria BTL (Blooms Taxonomy Level) from the questions framed.

1. Fundamental knowledge from Level-1 (Remember)-20%
2. Fundamental knowledge from Level- (Understand) : 30 %
3. Knowledge on application and analysis from Level-3(Apply) & 4 (Analyze) : 50 %